Repos - SIH 26147 on GitHub

Public GitHub repositories attempting *Automated model for analysis of .IQ and .wav files along with signal parameter extraction* (NTRO, SIH 2026), ranked strongest to weakest.

How this was built

- Searched GitHub (repo name, description, README and code search) for 26147 / SIH26147 / PS26147, the PS title, and topic keywords (IQ, wav, demodulation, de-interleaving, Viterbi, NTRO, SIH 2026), plus a web search. ~150 hits filtered to **43 relevant repos**; 16 more listed in the appendix as excluded.
- Nothing was cloned or run. Each repo was judged from its README, full file tree, languages and commit count via the GitHub API (snapshot 26 Sep 2026). Features are **as claimed** by the team; many READMEs are AI-written and optimistic.
- **Coverage** column maps to the PS: G = GUI, P = parameter identification (i), D = demodulation FSK/PSK/QAM (ii), I = de-interleaving block/conv/diagonal/PN (iii), F = FEC Viterbi/RS/concat/LDPC (iv), C = bit-stream correlation / header-payload (v). Y = present, P = partial, N = not evident.
- **Strength /10** weighs: coverage of i-v + GUI (50%), evidence it works - tests, ground-truth samples, real signals, commit history (30%), and depth of blind analysis vs. configured decoding (20%).

Landscape at a glance

- **15** repos score 7+ (claim the full i-v chain with real code), **14** are mid (5-7), **14** are early, narrow or off-target.
- Almost everyone does: file loader, PSD/waterfall/constellation, cumulant or ML modulation classifier, PSK/QAM/FSK demod, K=7 Viterbi + RS(255,223), CCSDS sync-word search. This is now **table stakes**.
- Rare (real differentiators): validation on **real over-the-air signals** (only ICHNOVA, Venkata-Manoj), **truly blind** interleaver size / PN-seed recovery via GF(2) rank (ShantanuSomwanshi, NaidhruvK, WaghJagdish), verification by CRC/re-encode instead of fake confidence (SIG-ID, Venkata-Manoj), **C++ performance** (arachknight66, Alekhkr, sanket), IQ+WAV fusion (SIGNALX).
- Weak spots across the field: blind LDPC (everyone uses a fixed catalogue), HF fading/equalisation, multi-signal captures, scramblers, and most GUIs are web dashboards rather than analyst desktop tools.
- Common style trend: 'epistemic honesty' - explicit UNKNOWN states, provenance labels, confidence gates. Judges will likely see this in many entries.

Ranked repositories

#	Repository	USP	Features	Solution vs PS (coverage)	Strength
1	ShantanuSomwanshi/SIH-Rf-Signal-Intelligence 2 commits, created 2026-09-22	Cleanest 1:1 mapping of every PS clause to a source file. Blindly identifies interleaver **including exact size / PN seed** and FEC type; 7 ground-truth sample files cover every combo.	WAV (stereo IQ / mono real), raw IQ cf32/ci16/ci8/cu8; BPSK, QPSK, 8PSK, 16QAM, 2/4/8-FSK; conv K=3/5/7 soft Viterbi, RS, RS+conv concatenated, LDPC min-sum; block/diagonal/convolutional/PN de-interleave; sync-word + blind bit-constancy header finder; GUI: spectrum, waterfall, constellation, eye, symbol-rate line.	Full chain file -> params -> demod -> de-interleave -> FEC -> header/payload. Offline pure Python. Fs for headerless IQ from filename hint or analyst. **G:Y P:Y D:Y I:Y F:Y C:Y**	**9.0/10** Most complete coverage of i-v with ground-truth samples; compact codebase (17 files, few commits).
2	NaidhruvK/wavSIH26 100+ commits, created 2026-09 (pushed 09-14)	Most rigorous team engineering: staged pipeline S0-S6 with contracts, a seeded signal 'zoo' corpus with truth JSON, and an eval harness that found and fixed real estimator biases.	Ranked-hypothesis raw IQ dtype sniffer; PSD/SNR/burst detect; receiver with carrier, timing, equaliser, lock-check, soft demap; GF(2) rank-collapse interleaver recovery, PN interleavers, phase-rotation search; conv/RS/LDPC catalogue decoders; descrambler, CCSDS framing, payload extraction; ML classifier; Docker.	Blind recovery chain down to CCSDS frames, validated against its own corpus with measured metrics. README is empty - no GUI evident. **G:N? P:Y D:Y I:Y F:Y C:Y**	**8.8/10** Deepest verified DSP + coding work (100 commits); weak on GUI/docs.
3	Gururaghavendra123/sihps2-2026 4 commits, created 2026-09-04	Answers are **verified by hard ground truth** (CRC-16, sync correlation, re-encode BER) instead of heuristic %; adaptive coarse-to-fine candidate search gives 4-5x speedup over brute force; says 'unknown' rather than false positives.	Oerder-Meyr symbol rate, cumulants C20/C40/C42; BPSK, QPSK, 8PSK, 16QAM, 2/4-FSK; M&M; timing, Costas; block/conv/diagonal/PN; Viterbi K=7, RS(255,223), concatenated, LDPC BP; PyQt6 desktop + FastAPI web dashboard; DEEPDIVE.md math doc.	Catalogue-bounded blind identification (6 mod x 4 FEC x 4 interleavers = 96 hypotheses) with verification gate, then payload extraction. **G:Y P:Y D:Y I:Y F:Y C:Y**	**8.6/10** Complete and honest; blind only within a fixed catalogue.
4	WaghJagdish/PS147 1 commits, created 2026-09-19	Hypothesis-driven **beam search** with composite confidence (AMC margin, GF(2) rank drop, FEC syndrome ratio) and aleatoric-vs-epistemic uncertainty labels; self-verifying payload states.	Raw IQ dtype selector with ambiguity warning; CFAR emission band overlay; PSD, waterfall, I/Q scope, constellation; HOC + ML classifier; block & conv de-interleavers via GF(2) rank; conv, RS, LDPC, FEC estimator; descrambler, frame sync; hex/ASCII payload viewer; in-UI synthetic generator; FastAPI + React.	End-to-end blind search over modulation/interleaver/FEC with explicit uncertainty; web workstation GUI. **G:Y P:Y D:Y I:P F:Y C:Y**	**8.5/10** Strong design & explainability; single bulk commit, diagonal/PN not evident.
5	yoganandasp15/SIH_2026_Signal-Analyser 2 commits, created 2026-09-26	'Physics first' epistemic engine: measure -> hypothesise -> search for contradictions -> validation gates; six explicit verdict states (VALIDATED, ...). Argues against black-box DL for SIGINT.	Preprocessing (DC, IQ imbalance, AGC, resampler); cumulant, cyclostationary, phase, spectral features; PSK/QAM/FSK demod with LLR soft output; block/conv/diagonal/PN de-interleave engine; concatenated, Viterbi, RS, LDPC; GUI app + CLI.	Autonomous blind parameter engine + candidate ranker + contradiction analyzer feeding full decode chain. **G:Y P:Y D:Y I:Y F:Y C:Y**	**8.4/10** Very broad and well argued (104 code files); only 2 commits, claims untested by us.
6	The-4Script/RadioFry 100 commits, created 2026-09-04	Heaviest **ML** story: trained CNN modulation model (RadioML 2018 + augmentation) plus separate trained **interleaver and FEC classifiers**; GNU Radio integration; 13-page Streamlit app.	Pages: upload, parameters, modulation, demod, de-interleaving, FEC, correlation, report, 'time machine', cyclostationary, capability, fusion landscape, 'signal DNA'; RadioML 2016/2018 + real-world multipath loaders; de-interleave search; bit-structure analysis.	Hybrid ML + DSP pipeline; missing model artefacts reported as unavailable instead of faked. **G:Y P:Y D:Y I:Y F:Y C:Y**	**8.2/10** Most mature ML (100 commits). forge-script/RadioFry is a copy of this repo.
7	Venkata-Manoj/RF-signal-analysis 51 commits, created 2026-09-25	FEC + interleaver reported **only when proven by a CRC check**; three interfaces (desktop, web, CLI); tested on real over-the-air recordings and benchmarked vs SNR.	BPSK, QPSK, 8PSK, 16/64-QAM, 2/4-FSK with confidence; Fs, BW, symbol rate, CFO, SNR, EVM/MER; sync-word header position; decoded text/hex/ASCII; 8-tab GUI (time, spectrum, waterfall, constellation, eye...); batch analysis; OTA eval + performance benchmarks.	Full pipeline with conservative 'verified' semantics and batch mode. **G:Y P:Y D:Y I:Y F:Y C:Y**	**8.0/10** Polished and user-friendly (51 commits); repo also carries unrelated UI-tooling scripts.
8	SomeNobody21112/ICHNOVA 75 commits, created 2026-09-16	Only repo **validated on real government transmissions** (WWV, DCF77, MSF, JJY, DDH47 RTTY, All India Radio via KiwiSDR) with GPS-checked results; statistical significance corrected for number of hypotheses tried.	Blind CFO, symbol rate, BPSK/QPSK, phase search; conv codes K=3/5/7 r=1/2 x block interleaver with dual-code syndrome test; outcomes DECODED / SIGNAL_NO_CODE / UNKNOWN; bench suites, null set, performance reports; React dashboard.	Narrow but deep: blind code+interleaver recovery with false-positive control and real-signal proof. **G:Y P:Y D:P I:P F:P C:Y**	**7.8/10** Best scientific credibility (75 commits); limited modulation/FEC range.
9	Rohanbhandari9703/SignalIQ 1 commits, created 2026-09-20	Classic analyst desktop tool: PySide6 + pyqtgraph 12-stage pipeline with a **brute-force FEC x de-interleaver grid search**; builds a standalone .exe.	WAV, raw IQ (cf32/cf64/ci16/ci8), SigMF; Welch PSD, spectrogram, BW; M2M4 SNR; CFO power-of-N; cyclostationary symbol rate; Random Forest AMC (92.5% CV) BPSK/QPSK/8PSK/16QAM/2FSK/4FSK; Costas, Gardner; Viterbi K=7, RS(255,223); CCSDS ASM / Barker sync; eye diagram, bit viewer; JSON report, session save.	Automatic 'Run Pipeline (F5)' from file to frames. **G:Y P:Y D:Y I:P F:P C:Y**	**7.6/10** Solid, focused; no LDPC, 1 commit.

#	Repository	USP	Features	Solution vs PS (coverage)	Strength
10	EDM-Fan/Nova-Signum---Signal-Analyzer- 1 commits, created 2026-09-22	Explicit **phase-ambiguity resolution** (0/90/180/270) during frame sync plus blind FEC auto-detection across RS/LDPC/concat/conv with 4 de-interleaver architectures.	Random Forest on cumulants/kurtosis (BPSK, QPSK, 16QAM, 2FSK); CFO, Gardner/M&M; sync words 0xEB902A3C / CCSDS; RS(64,48)/(32,24), LDPC(128,64), K=7 conv; PDF/JSON/CSV report; cyclostationary baud on load; PySide6 GUI; many demo generators.	Full chain with re-encode match metric and ground-truth BER on synthetic files. **G:Y P:Y D:Y I:Y F:Y C:Y**	**7.5/10** Complete but only 4 modulations; 1 commit.
11	rabishankar21/SignalSight 10 commits, created 2026-09-25	Desktop suite covering **every listed item** including 64QAM and LDPC(256,128) min-sum, with separate validation scripts per stage.	IQ/WAV ingest, preprocessing, PSD, waterfall, constellation, SNR, BW, symbol rate, CFO; BPSK/QPSK/8PSK/2FSK/16QAM/64QAM; Viterbi, RS, concat, LDPC; block/conv/diagonal/PN; interleaver & FEC detectors; bitstream correlation, frame-boundary detection.	Full chain in a desktop GUI. **G:Y P:Y D:Y I:Y F:Y C:Y**	**7.4/10** Feature-complete; many UI-patch scripts suggest GUI churn.
12	arachknight66/SIH26147 40 commits, created 2026-08-27	Only repo with a real **C++20 core + pybind11** (soft Viterbi K=7, RS, LDPC, correlation, CRC, PSD/STFT, receivers) and chunked streaming for big files.	6-layer forensic pipeline (ingest, stats, blind AMC, sync, de-interleave+FEC, frame recovery); epistemic gating; headless CLI + PySide6 GUI; C++ unit tests & benchmarks.	Performance-oriented pipeline; README admits native decoders are configured, not blind. **G:Y P:Y D:Y I:P F:Y C:Y**	**7.4/10** Best performance engineering (40 commits); blind coding-layer weak.
13	Krezmitent/SigNull 40 commits, created 2026-09-09	Same 'Signal Analysis MVP' 6-layer architecture as arachknight66 (shared origin) in pure Python, adding **OFDM** and vectorised CRC search.	Loaders, rate estimation, sync, demod, de-interleaving, conv/RS/LDPC/concat FEC, hypotheses, framing, correlation; CLI JSON + PySide6 GUI; stereo WAV real-vs-IQ prompt.	Full chain with explicit certainty gating. **G:Y P:Y D:Y I:Y F:Y C:Y**	**7.2/10** Broad (40 commits); overlaps heavily with arachknight66.
14	Alekhkr/PS2 4 commits, created 2026-09-23	Scientific-instrument style desktop app with **per-parameter provenance** (ParameterEvidence) and C++ fast kernels.	Canonical SignalBuffer ingest; DC/IQ-imbalance conditioning; synced time/PSD/waterfall/constellation; carrier, BW, SNR, symbol-rate candidates; hybrid classifier; FSK/BPSK/QPSK/8PSK/QAM16/64 demod; Viterbi, RS, LDPC, concat hypothesis engines; correlator.	Full chain in PySide6 + PyQtGraph. **G:Y P:Y D:Y I:P F:Y C:Y**	**7.0/10** Good structure; thin README, 4 commits.
15	harikesh2709-creator/spectra-signal-analyzer 4 commits, created 2026-09-25	Most **presentation-ready**: 5-min narrated demo video, decks, infographic; 60 FPS WebGL waterfall; team FUTURISTICS.	DDC, RRC, Gardner/Costas sync, AMC + trained model, baud & SNR estimators, blind interleaver detector, de-interleaver, Viterbi, RS, LDPC, concatenated, correlator, frame parser, generator; FastAPI + web UI.	Full chain claimed in an 'air-gapped SIGINT terminal' UI. **G:Y P:Y D:Y I:Y F:Y C:Y**	**7.0/10** Code covers everything but claims are marketing-heavy; 4 commits.
16	ashcodes1511/SIH-2026-Chroma-Code 9 commits, created 2026-09-12	Adds **cyber-forensics**: Shannon entropy to tell plaintext vs encrypted payload, plus a 1D-CNN modulation classifier.	Multi-SDR binary parser (RTL-SDR, HackRF, BladeRF, USRP); Hilbert for mono WAV; normaliser; heuristic + CNN AMC; RRC, Costas, M&M; PSK/QAM/FSK; de-interleaver; Viterbi, RS; sync detector; FastAPI dashboard.	Pipeline extends past the PS into payload forensics. **G:Y P:Y D:Y I:P F:P C:Y**	**6.9/10** Good breadth; no LDPC; 9 commits.
17	Devansh-567/hackathon 1 commits, created 2026-09-17	Most **honestly documented** backend: 188/188 tests, 5-state honesty vocabulary, 'what's not built yet' list; handles huge files via memory-mapped chunking.	WAV + raw IQ + full SigMF datatype parser; synthetic generator (7 mods, impairments); rule-based explainable classifier (M-th power); PSK/QAM/FSK demod with EVM/lock; conv+Viterbi (commpy), RS (reedsolo) with syndrome self-check; interleavers; JSON/CSV/PDF/SigMF reports; React UI with batch, compare, similar-signal fingerprinting; pitch deck.	Tested foundation; LDPC and some coding layers not complete. **G:Y P:Y D:Y I:P F:P C:P**	**6.9/10** High engineering quality, incomplete scope.
18	karurravishankermohit-stack/-SignalX 32 commits, created 2026-09-16	Strict **provenance labels** on every value (METADATA / USER / DSP_ESTIMATED / AUTO_CLASSIFIED / DEMO / UNAVAILABLE); 17-stage pipeline.	Parser, preprocessing, spectrum, waterfall, constellation, parameter estimator, modulation classifier, demod, interleave detector, FEC codec/detector (Viterbi, RS, concat, LDPC check), frame detector, correlation; FastAPI + React/Plotly; SQLite sessions; Firebase auth.	Full workstation with case management. **G:Y P:Y D:Y I:Y F:P C:Y**	**6.8/10** Broad (32 commits); web/auth overhead.
19	arunkumarmeda27/SpectraSync 19 commits, created 2026-09-18	Enterprise infra: auth, RBAC, Docker, Postgres/Redis, **WebSocket live stage progress**, forensic PDF reports, 6 golden demo vectors.	13-stage DSP pipeline; cumulant (C40/C42/C63) + Random Forest AMC incl AM/FM; carrier/clock recovery; de-interleave; FEC; bitstream analysis; cross-correlation; 8-panel neon dashboard; files up to 2 GB.	Full pipeline wrapped in production web stack. **G:Y P:Y D:Y I:P F:P C:Y**	**6.6/10** Infra outweighs DSP depth; no LDPC evident.
20	Andro-HM/IQWAV 39 commits, created 2026-08-29	Research-grade **experiment log**: head-to-head symbol-rate and timing benchmarks, held-out validation, blind occupied-band detection on real IQ, notebooks. (Mirror: iamsayandas2-blip/IQWAV.)	WAV / stereo-IQ / raw IQ ingest; activity & band extraction; PSD, spectrogram; BW, CFO, noise, symbol rate; rule-based AMC with dataset/eval; auto/cross-correlation; analog FM demod; impairments model.	Parameter estimation done carefully; FEC/de-interleaving still future work. **G:P P:Y D:P I:N F:N C:P**	**6.4/10** Best research discipline (39 commits); coding layer missing.

#	Repository	USP	Features	Solution vs PS (coverage)	Strength
21	aayushkasurde112-cell/RF_MVP 2 commits, created 2026-09-11	Strong DSP fundamentals: **raw-IQ dtype auto-guess** by file-size modulo + float plausibility probe, multi-signal CFAR channelisation, Farrow interpolator.	SigMF sidecar writer; Welch PSD + median/MAD CFAR; burst channelisation; delay-multiply/envelope baud lines; HOC C40/C42/C63 decision tree; M-th power CFO, Gardner+Farrow, Costas, FSK discriminator; later phases: demod, interleaver, FEC, bitstream, FastAPI server.	Stateless library pipeline designed to run per-chunk in a streaming backend. **G:N P:Y D:Y I:P F:P C:P**	**6.3/10** Good core; GUI not built.
22	Jagzz-Coder/sigscope 1 commits, created 2026-09-20	Runs **100% in the browser** (pure JS, no backend): blind convolutional-code identification + Viterbi and GF(2) rank-sweep structure detection.	Symbol rate, CFO, 7 modulations, demod, sync-word/frame correlation, outer block code; 38 passing Node tests over a full concatenated chain.	Blind chain in a single JS core file. **G:Y P:Y D:Y I:P F:P C:Y**	**6.2/10** Clever and tested; tiny codebase, no LDPC.
23	Priyanshugrawal/AutoSIG 1 commits, created 2026-09-19	Guided 8-stage Streamlit dashboard with a **recovery validation** stage (CRC/parity, overall recovery score).	Validation, DC/IQ correction, burst detection, PSD/waterfall/constellation, BW/SNR/symbol rate/CFO; ML AMC (FSK..64QAM) + ranked FEC/interleaver hypotheses; demod, de-interleave, Viterbi/RS/concat/LDPC-heuristic; header/payload parse; report.	Full chain, hypothesis chosen from ML ranking. **G:Y P:Y D:Y I:Y F:P C:Y**	**6.0/10** Covers list; LDPC heuristic; 1 commit, 14 files.
24	tejaswi-jain2007/SIGNEX 5 commits, created 2026-09-18	**ResNet-18 AMC** over 10+ classes incl 16/32APSK and OFDM; air-gapped design, audit logs, batch queue.	FastAPI + React; time/spectrum/spectrogram/constellation; hex/ASCII bitstream; interleaver, FEC, correlator, text decoder modules; PDF/JSON dossiers.	DL-first classification with decode modules attached. **G:Y P:Y D:Y I:P F:P C:Y**	**5.9/10** Depth of coding layer unclear.
25	RohithvijayR/SIGNALX 6 commits, created 2026-09-10	Only repo doing **IQ + WAV multi-modal fusion** (paired files, IQ-only vs WAV-only vs fused predictions) - directly echoes the PS line about training data 'containing both .IQ and .wav'.	Parsers for IQ (several dtypes), WAV, SigMF; time/FFT/Welch/STFT/constellation; SNR/BW; ML classifier; FEC and interleaver *detectors*; ground-truth comparison table; React dashboard; history DB.	Profiles the signal; decoding not evident. **G:Y P:Y D:N I:P F:P C:N**	**5.8/10** Unique fusion idea; no demod/decoding.
26	GaurangJagtap/Radio-Frequency-Signal-Analyzer 1 commits, created 2026-09-09	Explainable-RF 'math trace' and cascade gate; even a voice controller.	Cyclostationary, multi-domain features, ML classifier, demodulators, de-interleavers, FEC decoders incl LDPC & concatenated, bitstream correlator, payload extractor; SDR-style dark GUI.	README still says 'prototype with mock data' while code tree has real modules. **G:Y P:Y D:Y I:Y F:Y C:Y**	**5.6/10** Inconsistent docs vs code; 1 commit.
27	anshumanarchit-crypto/machine-learning-SIH 16 commits, created 2026-09-25	Evidence-first **decision layer**: rule vs ML (Random Forest) consensus, calibrated abstention to UNKNOWN, evidence ledger, replay engine.	Cumulants C20..C80, M2M4 SNR, EVM, cluster metrics; threshold sweeps & calibration; Streamlit GUI reading result.json.	One member's slice of a team pipeline; demod/FEC in other repos. **G:Y P:Y D:N I:N F:N C:N**	**5.4/10** Good confidence engineering, partial by design.
28	VASANTH-Balaji/signal-analyzer-webapp 4 commits, created 2026-09-23	Streamlit **public web deployment** of a desktop analyzer's backend.	Blind detect, symbol-rate estimate, modulation classifier, BPSK + general demod, FEC core, RS, LDPC, bitstream correlation; sample .iq files.	Full-ish chain exposed via web. **G:Y P:Y D:Y I:N F:P C:Y**	**5.3/10** Thin docs, 13 files.
29	NirmitSingh-main/SYNAPS 38 commits, created 2026-08-26	Heavy AI: CNN, Transformer and **self-supervised MAE** models, unknown-signal detection, LLM 'copilot'. (Forks: SiddhantMenon14, pratyushh2.)	IQ/WAV loaders, spectrogram/constellation, training & eval, confidence, analysis/report APIs, React frontend, Supabase.	Classification + assistant; coding layer limited. **G:Y P:Y D:P I:P F:N C:P**	**5.0/10** Ambitious ML, minimal README (38 commits).
30	pranshu1141-sharma/Pinpoint 31 commits, created 2026-09-11	Exceptional documentation: requirement-by-requirement status, **judge-defense answers**, 1 GiB processing evidence.	Time/frequency region detection, pulse timing, isolation layers, coarse envelope + BPSK/QPSK classification, symbol rate, SigMF, calibration corpus; React dashboard.	Explicitly **no demodulation or decoding**. **G:Y P:Y D:N I:N F:N C:N**	**4.9/10** Honest but covers only task (i) partly.
31	gourabde7/Spectra 12 commits, created 2026-08-28	Smallest end-to-end demo: one GUI with 6 tabs + 'Run Full Pipeline' button.	Loader (cf32/ci16/cu8); spectrum, waterfall, constellation; BPSK/QPSK/16QAM/FSK; 4 de-interleavers; Viterbi K=7, RS, LDPC, concat; preamble search; demo signal generator; architecture slide.	Self-described ideation-round prototype using heuristics. **G:Y P:P D:Y I:Y F:P C:Y**	**4.8/10** Breadth with shallow depth (8 files).
32	sautiksamui-tech/SIH2026 1 commits, created 2026-09-08	Simple PySide6 app with demo mode.	WAV/raw IQ; PSD, OBW, SNR, CFO; evidence-based modulation candidates; Costas BPSK/QPSK; interleaver search + basic FEC framework; tests.	Timing/FEC partly stubs. **G:Y P:Y D:P I:P F:P C:N**	**4.3/10** Early-stage.

#	Repository	USP	Features	Solution vs PS (coverage)	Strength
33	Akshat030307/SIH 3 commits, created 2026-08-31	Focus on **multi-signal captures**: per-signal boxes on spectrogram, SigMF annotations, broad classes (AM/FM/SSB/CW/FSK/PSK/QAM/OFDM/chirp).	RadioML 2016 loader, file I/O with dtype guesser and filename parsers; spectrogram, noise floor, detector (WIP); CNN planned.	Phase 3 of 8 in progress; no demod/FEC yet. **G:N P:P D:N I:N F:N C:N**	**4.2/10** Good plan, early.
34	PradhanaService/sanket 2 commits, created 2026-09-15	**C++20 / Qt6** high-performance viewer: zero-copy mmap of 50 GB+ files, AVX2 SIMD, FFTW waterfall; GNU Radio C++ signal generator.	IQ/WAV readers, waterfall, constellation, time plot, feature extractor, playback controls.	Visualisation layer only. **G:Y P:P D:N I:N F:N C:N**	**4.1/10** Great performance, tiny scope.
35	hn260/signalscope 2 commits, created 2026-09-25	Offline Streamlit analyzer with an explicit **honesty policy** (relative vs absolute frequency, two BW methods).	IQ/WAV ingest, waveform/spectrum/spectrogram, params with confidence, statistical modulation classifier, JSON/CSV reports, tests.	Parameters + classification only. **G:Y P:Y D:N I:N F:N C:N**	**3.8/10** Clean but narrow.
36	kunal-shetty/SIH2026_Signal_Intelligence_Recognition_Engine 28 commits, created 2026-09-10	'**Shazam for radio signals**': RadioML 2016 CNN, 11 classes, < 2 s per file, batch of 50.	IQ/WAV parse, centre freq/BW/SNR/symbol rate, spectrogram/PSD/waveform, FastAPI + React, Docker, synthetic test endpoint.	Classification + parameters only. **G:Y P:Y D:N I:N F:N C:N**	**3.8/10** Covers only task (i) partly.
37	DevWithShubham18/SAGE-RF 46 commits, created 2026-09-07	Actually **deployed** (Vercel + Render) with Firebase auth and an LLM RF assistant.	Spectral analysis, waterfall, signal detection, modulation estimation, Fourier/Laplace views, history, PDF reports.	General RF workstation; no demod/FEC. **G:Y P:P D:N I:N F:N C:N**	**3.6/10** Off the PS core.
38	vedantpanchal77/Team-Overflow 2 commits, created 2026-09-12	DSP metrics + **Gemini-generated** diagnostic reports with heuristic fallback.	IQ (cf32/cs16/cu8) and WAV; time, frequency and RF metrics (IQ imbalance, OBW, spectral flatness); candidate modulation; Plotly quad view.	Measurement dashboard only. **G:Y P:Y D:N I:N F:N C:N**	**3.4/10** Narrow; LLM reliance.
39	Selvamurugan-hub/SpectraSense-AI 4 commits, created 2026-09-09	Signal **fingerprinting**, historical baseline comparison and anomaly detection.	IQ/WAV loader, segmentation, parameter extraction, health score, 'what changed' table, PDF reports, React UI.	Reframes PS as monitoring/anomaly - misses demod/FEC. **G:Y P:P D:N I:N F:N C:N**	**3.3/10** Misaligned with PS asks.
40	SrishtiPriya27/signalscope 3 commits, created 2026-09-16	Polished React/Vite **frontend-only** workstation (Vercel-ready).	Pages for every stage (spectrum, constellation, demod, de-interleave, FEC, correlation, report) but results come from a MockSignalProcessingClient.	UI mock-up; processing is simulated. **G:Y P:N D:N I:N F:N C:N**	**2.8/10** Good UI reference only.
41	adityabisoyee28-cpu/SignalLens-AI 22 commits, created 2026-08-26	Supabase-backed web app with Random Forest over 6 toy classes (sine, AM, FM, PM...).	WAV/IQ (RTL-SDR uint8, complex64); time/FFT/PSD/spectrogram/constellation; 11-dim features.	Classifies analog toy signals - not the digital mods the PS needs. **G:Y P:P D:N I:N F:N C:N**	**2.6/10** Weak fit.
42	Muizzahmed786/SIH26147 ? commits, created 2026-09 (pushed 09-18)	Planning only: 'PROJECT_BRAIN' markdown set (architecture, DSP, FEC & interleaving, ML, MVP scope, phases).	Documentation, no code.	Design notes. **G:N P:N D:N I:N F:N C:N**	**1.8/10** Useful reading, nothing built.
43	xarjunpatil/SIH26147-Automated-model-for-analysis-of-IQ-and-wav-files-along-with-signal 1 commits, created 2026-08-31	Repo named after the PS with full PS text in README.	Single app.py + test, Dockerfile.	Skeleton. **G:P P:N D:N I:N F:N C:N**	**1.5/10** Placeholder.

Appendix - found but excluded

Repository	Why excluded
forge-script/RadioFry	Copy of The-4Script/RadioFry (same README/code, 1 commit).
SiddhantMenon14/SYNAPS, pratyushh2/SYNAPS	Forks of NirmitSingh-main/SYNAPS.
iamsayandas2-blip/IQWAV	Mirror of Andro-HM/IQWAV (same project/README).
harikesh2709-creator/FUTURISTICS	Different problem (ocean freight forecasting), despite search hits.
harshgounder/sih-2026	Different problem (SIH26166).
mks-Roald/ps26147_toolkit, iparayush/SignalLens-AI-	Indexed on the web but now 404 (deleted or private).
harideskzone-hue/SIH26147-Border	Empty repository.
AryanSahu321/sih, jeevansai-hub/SIH-2026-, Vigneshrdy/sih-ps-archive, Rajkumar-Porandla/SIH-2026-Problem-Statements, NoBugNinja/Smart-India-Hackathon-SIH-2026-Problem-Statements, Abhinav-Prabhakar/SIH2026, Ardcode12/SIH26soft	Lists / archives of all SIH problem statements, not solutions.

Search limits: GitHub search only indexes public repos and some READMEs lag; private or newly-created repos, or ones without the PS number/title in name, description or README, can be missed. Scores are a desk judgement from READMEs and file trees, not from running the code.